

Test 4

April 30, 2024

Problem 1

A barn has dimensions 30×40 ft. A cow is tethered to a corner of the barn by a rope 70 ft long. What is the locus of points where the cow may graze?

Problem 2

A circle has a radius of 4 meters. What is the locus of points whose distance is 2 meters away from the circle?

Problem 3

A cylinder has a radius of 6 in and a height of 12 in. Find a) its surface area and b) its volume.

Problem 4

A sphere has a radius of 6 ft. Find a) its surface area and b) its volume.

Problem 5

A snowman is composed of three spherical snowballs and a cone carrot for the nose. If each of the snowballs has a radius of 10 in, and the carrot is 8 in long with a base radius of 1 in, what is a) the volume of the snowman and b) the surface area of the snowman?

Problem 6

Calculate the distance between the following sets of points:

a) $(0, -3)$ and $(0, 9)$

b) $(0, 0)$ and $(3, 4)$

c) $(-2, -9)$ and $(9, 3)$

d) $(10, -3)$ and $(4, 5)$

Problem 7

Find the equation of a line traveling through the points:

a) $(0, -3)$ and $(4, 5)$

b) $(0, 0)$ and $(3, 4)$

c) $(-2, -9)$ and $(6, -1)$

d) $(10, -3)$ and $(4, 5)$

Problem 8

Solve the right triangle with a leg length of 5, and whose hypotenuse has a length 13.

Problem 9

Calculate the following:

a) $\sin\left(\frac{\pi}{2}\right)$

b) $\cos\left(\frac{5\pi}{6}\right)$

c) $\sec\left(\frac{7\pi}{6}\right)$

d) $\tan\left(\frac{\pi}{4}\right)$

Extra Credit

Prove that the angle sum of any triangle is a straight angle. (Hint: you will need to use the parallel postulate; if you're not using it, you likely have an invalid proof)